

12. Test cases

The test case are run with https://github.com/bsanderse/SDE-sandbox/blob/main/src/SDE_samplers/Langevin_PFF_nD.m, commit 1826128.

The runs are reproducible by using `rng(10)`. The density plots are made with Matlab's `ksdensity` function with 100 points.

12.1. A: Nonlinear model

This test case is adapted from Evensen chapter 18 [21] – we have removed the observation error and fixed the measurement. We aim to sample from

$$\rho(\mathbf{u}) := p(\mathbf{u}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{u})p(\mathbf{u}), \quad (85)$$

with the likelihood

$$p(\mathbf{y}|\mathbf{u}) = \exp\left(-\frac{1}{2}\|\mathbf{y} - \mathbf{H}(\mathbf{u})\|_{\mathbf{R}}^2\right), \quad (86)$$

and prior

$$p(\mathbf{u}) = \exp\left(-\frac{1}{2}\|\mathbf{u} - \boldsymbol{\mu}_0\|_{\mathbf{B}}^2\right). \quad (87)$$

We take

$$\mathbf{H}(\mathbf{u}) = 1 + \sin(\pi\mathbf{u}), \quad (88)$$

$\mathbf{y} = 0$, $\mathbf{R} = 1$, $\boldsymbol{\mu}_0 = 1$, $\mathbf{B} = 1$.

The initial condition (e.g. for particle flow or SDE) is sampled from the prior.

Method	M	kernel	time integration	$\Delta\tau$	T	burn-in
Direct	200	-	-	-	-	-
Langevin (SDE)	200	-	Euler-Maruyama	1/100	100	$T/4$
Particle flow filter (ODE)	200	π	Forward Euler	1/100	100	-

Table 1: Settings for test case A

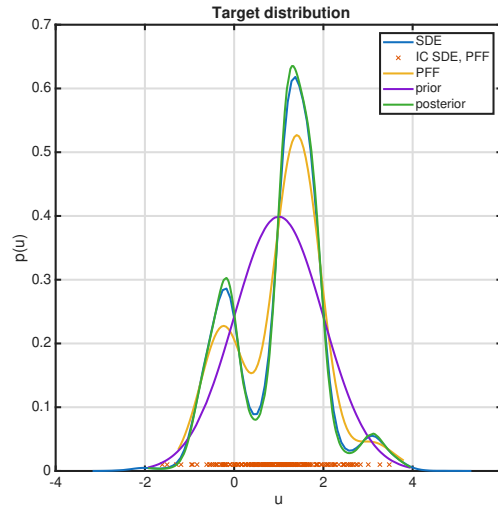


Figure 7: Results for testcase A. IC = initial condition.

12.2. B: Sampling a Gaussian mixture in n -D

In this test case the aim is to sample from a D -dimensional Gaussian mixture, consisting of K terms:

$$\rho(\mathbf{u}) = \sum_{i=1}^K c_i \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i). \quad (89)$$

This is not a posterior, but all sampling methods can be straightforwardly applied. See for example <https://akorba.github.io/resources/teaching/Bayesian.pdf>, slide 25. We take $K = 8$, $c_i = \frac{1}{K}$, $\boldsymbol{\mu}_i \sim \mathcal{U}(-L, L)^D$, $\boldsymbol{\Sigma}_i = \mathbf{I}_D$ and $L = 40$.

The initial condition has a strong effect on whether all modes of the mixture get sampled. Korba takes $\mathbf{u} \sim \mathcal{N}(0, \mathbf{I})$, but this leads to poor results. A better choice is $\mathbf{u} \sim \mathcal{U}[-L, L]^D$.

Method	M	kernel	time integration	$\Delta\tau$	T	burn-in
Direct	200	-	-	-	-	-
Langevin (SDE)	200	-	Euler-Maruyama	1/100	100	$T/4$
Particle flow filter (ODE)	200	π	Forward Euler	1/100	100	-

Table 2: Settings for test case B

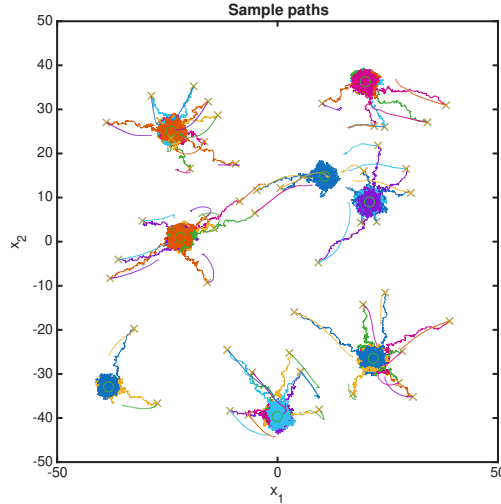


Figure 8: Results for testcase B. Green circles indicate the centers of the Gaussian mixtures.

12.3. C: Sampling a Gaussian mixture - 1D

Liu and Wang [10] discuss a particular example of 12.2 where $D = 1$, and they set $K = 2$, $c_1 = \frac{1}{3}$, $c_2 = \frac{2}{3}$, $\mu_1 = -2$, $\mu_2 = 2$, $\Sigma_1 = \Sigma_2 = 1$. Their initial condition is $\mathbf{u} \sim \mathcal{N}(-10, 1)$, which is chosen to be far from the modes of the mixture, so that there is basically no overlap between prior and target, and particles have to travel significantly.

Method	M	kernel	time integration	$\Delta\tau$	T	burn-in
Direct	200	-	-	-	-	-
Langevin (SDE)	200	-	Euler-Maruyama	1/100	500	$T/4$
Particle flow filter (ODE)	200	π	Forward Euler	1/100	500	-

Table 3: Settings for test case C

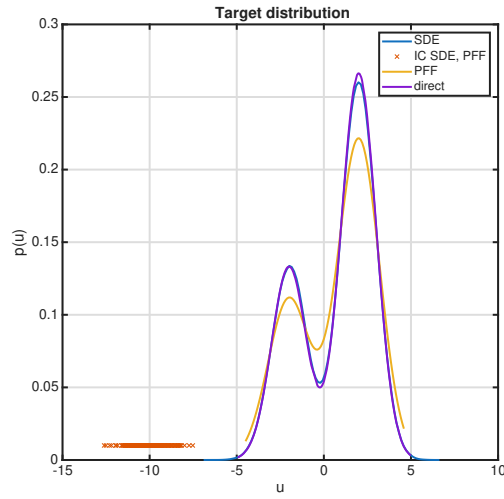


Figure 9: Results for testcase C. IC = initial condition.